Sharp growth of vertex-position ratios and realizable pairs

Haobo Ma
ChronoAI Pte Ltd; The Omega Institute

Wenlin Zhang
National University of Singapore; The Omega Institute

13 October 2026

Abstract

We prove that the largest ratio of the upper to lower vertex-position number among connected cubic graphs of order n is $\Theta(n/\log n)$, for all sufficiently large even n. For connected graphs of maximum degree exactly Δ, the corresponding maximum is $\Theta(n/(\Delta + \log n))$, uniformly for $3 \le \Delta \le n-1$. The proof extends a shortest-path-cover argument to geodesic antichains and constructs cubic graphs with linearly many incomparable vertices and logarithmically many root geodesics covering the graph. We also classify all pairs $(\mathrm{vp}^-, \mathrm{vp})$ with lower parameter at most two: they are $(1,1), (1,2)$ and $(2,b)$ for $2 \le b \le 10$; the largest upper parameter in row a grows exponentially in a. Finally, vertex-visibility ratios grow linearly along a cubic family, and local monophonic-position ratios are unbounded.

1 Vertex position and the main results

All graphs are finite, simple, undirected and connected. Following Thankachy et al. [1], an x-position set is a set $S \subseteq V(G)$ such that, for each $v \in S$, no vertex of $S \setminus \{v\}$ lies on any shortest x, v-path. Write $p_x(G)$ for its maximum size, and put

$$a = \mathrm{vp}^-(G) = \min_x p_x(G), \qquad b = \mathrm{vp}(G) = \max_x p_x(G), \qquad \rho(G) = b/a.$$

For $|V(G)| \ge 2$ a maximum set can be chosen to exclude x; a set containing x has size at most one. We use $p_x(K_1) = 1$.

Problem 2.4 of [1] asks whether ρ is bounded over connected graphs. James Tuite asked in addition about bounded degree, local path variants, and realizable pairs (private communication, 11 October 2026). The following results give the sharp order of growth and settle the first two rows of the pair problem. All logarithms in estimates below have base two.

Theorem 1. *There are absolute positive constants bounding the following maxima above and below by constant multiples of the displayed quantities:*

$$\begin{aligned}
\max_{|V(G)|=n,\ G\ \text{cubic}} \rho(G) &= \Theta\left(\frac{n}{\log n}\right), \quad n \text{ even and sufficiently large},\\
\max_{|V(G)|=n,\ \Delta(G)\le\Delta} \rho(G) &= \Theta\left(\frac{n}{\log n}\right), \quad \Delta \ge 3,\ n \text{ sufficiently large},\\
\max_{|V(G)|=n,\ \Delta(G)=\Delta} \rho(G) &= \Theta\left(\frac{n}{\Delta + \log n}\right), \quad 3 \le \Delta \le n-1.
\end{aligned}$$

In the last line the constants and the threshold for n are independent of Δ. If $a \le 2$, the realizable pairs (a, b) are exactly

$$(1,1), \quad (1,2), \quad (2,2), (2,3), \ldots, (2,10).$$

For every $a \ge 2$, the maximum $B(a) = \max\{\mathrm{vp}(G) : \mathrm{vp}^-(G) = a\}$ exists and satisfies

$$2^{a-1} \le B(a) \le (2a+1)4^a. \tag{1}$$

For maximum degree at most two, paths and cycles give a maximum ratio of two once $n \ge 3$. Trees have ratio at most two: if a tree has $\ell \ge 2$ leaves, then $p_x = \ell - 1$ at a leaf and $p_x = \ell$ elsewhere, apart from K_2 [1, Corollary 3.6]. The constructions below therefore require cycles even though their low-degree branching comes from trees.

2 An upper bound from geodesic covers

Fix x and define the geodesic order on $V(G) \setminus \{x\}$ by

$$u \preceq_x v \quad \Longleftrightarrow \quad d(x,u) + d(u,v) = d(x,v).$$

The triangle inequality proves transitivity, and strict order increases distance from x, so this is a partial order. Its antichains are exactly the observer-excluding x-position sets. A sphere about x is one such set.

Lemma 2. *If $p_y(G) = k$, then $V(G)$ is covered by k geodesics starting at y. Conversely, a cover by k such geodesics implies $p_y(G) \le k$.*

Proof. Dilworth's theorem [4] partitions the nonroot vertices into k chains. For a chain $v_1 \prec_y \cdots \prec_y v_s$, concatenate a shortest y, v_1-path and shortest paths between successive chain vertices. The total length telescopes to $d(y, v_s)$, so the concatenation is a geodesic containing the chain. Conversely, a position set meets each rooted geodesic in at most one vertex. □

We need a bound on all antichains, including those occupying several spheres. The following proof gives the endpointwise extension of the vertex-cover construction of Dumas–Foucaud–Perez–Todinca [2, Section 4, proof of Lemma 8]. Their recursion bounds spheres; the final slack argument below replaces the equal-radius condition.

Lemma 3. *If $V(G)$ is covered by k isometric paths, then*

$$p_x(G) \le (2k+1)4^k \qquad \text{for every } x.$$

Proof. Orient the covering paths $\mu_1, \ldots, \mu_k$ and call their indices colors. A vertex may receive the color of any base path containing it. Call a colored walk good if each color occurs in one block, following its base path monotonically. Every pair of vertices has a good shortest path: among colored shortest paths minimize the number of blocks; if a color occurs twice, replace the intervening part by the isometric base segment and color it uniformly, contradicting minimality. A block has sign $+$ or $-$ according to its direction; assign $+$ to a singleton block.

Take good shortest x, v-paths for v in an x-position set S. We describe the recursive grouping used in [2], recording lengths separately for each endpoint. At each node there is one current good walk to each endpoint in its group. Stop if all these walks are colored prefixes of one of them. Initially let C be the colors assigned to x. In each walk take the last vertex w colored in C, of color c, and replace the prefix to w by $\mu_c[x, w]$. Group the resulting walks by (c, sign). This first replacement does not increase length.

At a subsequent nonterminal node, take a longest colored prefix R such that every walk either extends R or is a prefix of R. Let z be its last vertex and set aside the walks ending within R. Call the others unfinished. Let C be the colors immediately after z. In each unfinished walk take the last vertex w whose color c lies in C. Replace its part after z through w by one edge from z to a vertex u of μ_c, followed by $\mu_c[u, w]$. If $z \in \mu_c$, take u toward w along that path. Otherwise choose the first neighbor of z in the fixed orientation of μ_c. Such a neighbor exists because $c \in C$. Group by the color and direction of this replacement. If the new block is a singleton and the color of z is also c, use the direction of zu. Assign the previously ended prefixes to any one nonempty group.

Here are the invariants that control this recursion. The last-occurrence choice removes all other colors in C from the continuing suffix. If c occurred in the retained prefix, it was its last color, so the replacement still has one monotone block of each color. Walks in the same group have the same successor of z. Thus their retained prefix advances at the next nonterminal node. After the first grouping, the active suffixes have a common initial color and direction. At a split with $|C| \geq 2$, the chosen child loses all colors of C except its chosen color. At a split with $C = \{c\}$, the previous initial color is different from c: otherwise all continuing walks would share the next base-path edge, contrary to maximality of the common prefix. That previous color has been passed and cannot recur. Consequently the union of available suffix colors strictly decreases at every subsequent split. There are at most k grouping steps from the root to a terminal node.

Children have distinct labels in the alphabet of $2k$ color–sign pairs. In any child, all the other sibling labels are forbidden in its descendants: the other colors have been removed; the opposite sign of its own color cannot recur in a good walk. These statements also hold for ended prefixes, which are carried unchanged and never create a branch.

Every replacement adds at most two edges. This is immediate if $z \in \mu_c$. Otherwise isometry gives

$$|\mu_c[u, w]| = d(u, w) \leq 1 + |W[z, w]|, \qquad 1 + |\mu_c[u, w]| \leq |W[z, w]| + 2,$$

where W is the walk being modified. Thus at recursion depth j the walk to endpoint v has length at most $d(x, v) + 2j$. This estimate uses no common endpoint radius. It is also valid if replacement walks repeat vertices: the coloring refers to occurrences and the retained-prefix argument applies to walks.

The number of terminal groups is at most 4^k. More generally, with q available labels there are at most 2^q terminal descendants. Induct on the remaining tree height. At a node with $t \geq 2$ children each child has at most $q - t + 1$ available labels, and

$$t\, 2^{q-t+1} \leq 2^q.$$

A unary node preserves the bound. Initially $q = 2k$.

In one terminal group all endpoint walks are prefixes of a common walk W. Write ℓ_v for the prefix length and $e_v = \ell_v - d(x, v)$ for its slack. Then $0 \leq e_v \leq 2k$. If u precedes v on W, the triangle inequality gives $e_u \leq e_v$. If equality holds, then

$$d(x, v) - d(x, u) = \ell_v - \ell_u \geq d(u, v) \geq d(x, v) - d(x, u),$$

so $u \preceq_x v$, contrary to the antichain condition. The integer slacks therefore strictly increase along the endpoints. Each terminal group has at most $2k+1$ endpoints, proving the result. □

Corollary 4. *Every connected graph of order $n \geq 2$ satisfies*

$$b \leq (2a + 1)4^a, \qquad \rho(G) \leq \frac{4n}{\log n}. \tag{2}$$

Proof. Choose y with $p_y = a$ and apply Lemmas 2 and 3. Hence

$$\rho \le \min\{(n-1)/a,\ 3 \cdot 4^a\}.$$

For $a \ge \frac{1}{4}\log n$ use the first term. Otherwise the second is less than $3\sqrt{n} \le 4n/\log n$. The last inequality follows, for example, by maximizing $(\log_2 t)/\sqrt{t}$ over positive real t, whose maximum is $2/(e \ln 2) < 4/3$. □

3 Cubic graphs attaining the growth order

We first build a subcubic graph with many incomparable targets. Its dyadic part is related to the exponential-sphere construction in [2, Figure 7]; a periodic binary-tree arrangement replicates its targets across enough rays to make their number proportional to the order.

3.1 The base graph

Fix $h \ge 8$, let $q = 2^h$ and $d = \lfloor \log h \rfloor$, and take a full binary tree T of depth d, rooted at r. Write

$$\ell = 2^d, \quad R = 2^{d+1} - 1, \quad w = h + R,$$

for its number of leaves, number of vertices, and the required number of rays. Choose a proper edge coloring $\chi : E(T) \to \{1, 2, 3\}$; recursively assign the two child edges colors different from the parent-edge color. Put

$$A = 4(q-1), \qquad M = 3d + 6, \qquad F = A + h, \qquad K = F + M + 1,$$
$$t_0 = 8\left\lceil \frac{h + M + 2}{8} \right\rceil, \qquad s = A + t_0, \qquad L = t_0 + 8q + 8M.$$

Let k be the least power of two at least w. Take a full binary hub tree with k leaves and root y. Attach a length-L ray to each of w distinct hub leaves, indexed from level 1 beyond its hub leaf. Name the rays $P_0, \dots, P_{h-1}$ and U_u for $u \in V(T)$, with $P_h = U_r$. Define

$$\delta_i = 4 \cdot 2^{h-i}, \qquad C_0 = \{s\}, \qquad C_i = \{c - \delta_i, c + \delta_i : c \in C_{i-1}\}.$$

The last set is

$$C_h = \{t_0 + 8j : 0 \le j < q\}. \tag{3}$$

Add precisely the edges

$$P_{i-1,t}P_{i,t} \quad (1 \le i \le h,\ t \in C_i), \tag{4}$$
$$U_{u,t}U_{v,t} \quad (uv \in E(T),\ 1 \le t \le L,\ t - t_0 \equiv \chi(uv) \pmod 8). \tag{5}$$

Call this graph $B^{(h)}$ and choose $x = P_{0,s}$.

The sets C_i and C_{i+1} are disjoint: their elements have different residues modulo $2\delta_{i+1}$. Thus an intermediate generator ray receives at most one extra edge at each vertex. At U_r the incoming dyadic edges have residue 0 modulo 8, whereas the tree edges have residues $1, 2, 3$. Proper edge coloring gives the same separation at every other U-ray. The added edges avoid the ray endpoints, since $t_0 \ge 2$ and $L \equiv 0 \pmod 8$ lies beyond all dyadic centers. Hence the maximum degree is three, the root has degree two, and there are exactly k leaves, including unused hub leaves.

Every added edge preserves ray level. The hub-root paths, extended along their attached rays, remain geodesics and cover the graph. The k leaves form a y-position set, so

$$p_y(B^{(h)}) = k, \qquad N_h = |V(B^{(h)})| = 2k - 1 + wL = \Theta(h2^h). \tag{6}$$

Here $h/2 \le \ell \le h$, $w = \Theta(h)$, $k = \Theta(h)$ and $L = \Theta(2^h)$.

3.2 Distances and the target antichain

Let H be the subgraph formed by the U-rays and (5), and let $C = \{U_{r,t} : t \in C_h\}$. Put

$$F_i = \sum_{j=1}^{i} (\delta_j + 1), \qquad F_0 = 0, \qquad F_h = F.$$

Define the potential

$$f(v) = \begin{cases} K, & v \text{ in the hub tree}, \\ \min\{K, F_i + \operatorname{dist}(t, C_i)\}, & v = P_{i,t},\ 0 \le i < h, \\ \min\{K, F + d_H(C, v)\}, & v \in V(H). \end{cases} \tag{7}$$

It is 1-Lipschitz on every edge. On rays and within H this follows from distance-to-a-set inequalities. On the root ray of H, $d_H(C, U_{r,t}) = \operatorname{dist}(t, C_h)$, since the level coordinate is 1-Lipschitz and the root ray realizes this lower bound. At a dyadic edge at level $t \in C_i$, the nearest point of C_{i-1} is at distance δ_i; the untruncated values on its two endpoints are $F_i - 1$ and F_i. To see the nearest-center assertion, centers in C_{i-1} are separated by at least $4\delta_i$, while each new center is δ_i from its parent. Finally, the first vertex of generator ray i has untruncated value $s + i - 1 \ge K$, and the first vertex of a ray of H has value at least $F + t_0 - 1 \ge K$. Thus edges to the hub also satisfy the bound.

Since $f(x) = 0$, distances from x are at least f. Inductively follow a ray by δ_i steps from a parent center and cross one dyadic edge. This gives length F_i paths to all centers of C_i, and therefore

$$d(x, P_{i,t}) = F_i \quad (t \in C_i), \qquad d(x, c) = F \quad (c \in C). \tag{8}$$

Consider the finite quotient $Q = V(T) \times \mathbb{Z}_8$, with a cycle on each u-fiber and the edge $(u, \chi(uv))(v, \chi(uv))$ for each $uv \in E(T)$. For a leaf u of T let $c_u = d_Q((r, 0), (u, 0))$. Following its tree path, using at most three cycle steps initially and finally and at most two between successive edge colors, gives

$$c_u \le 3 + d + 2(d - 1) + 3 = 3d + 4 < M.$$

Set $m_0 = \lceil M/8 \rceil$ and $J = \{m_0, \dots, q - 1 - m_0\}$. Then

$$d_H(C, U_{u,t_0+8j}) = c_u \qquad (u \text{ a leaf},\ j \in J). \tag{9}$$

Projection to Q proves the lower bound. For the upper bound, lift a shortest quotient path backwards from the indicated target. Its length is less than M, so every lifted level differs from the target level by less than M. The root-ray endpoint has residue 0 and, by the margins defining J, lies between t_0 and $t_0 + 8(q - 1)$; hence it belongs to C. The same margins and the padding in L keep the entire lift inside H.

Combining (7)–(9) yields

$$d(x, U_{u,t_0+8j}) = F + c_u < K.$$

Let S consist of all these targets. Targets on the same leaf ray have equal x-distance. A geodesic to a target cannot meet the hub, where the potential is K. Away from the hub, any visit to a different leaf ray must enter and leave via its parent ray. If the entry and exit levels are t, t', the excursion has length at least $|t - t'| + 2$ and can be shortened to the parent-ray segment of length $|t - t'|$. Thus such a visit cannot occur in a geodesic. It follows that S is an x-position set.

For $h \ge 8$, $m_0 \le (3 \log h + 13)/8 \le 2^{h-2}$: the second inequality holds at $h = 8$ and its right side doubles at each step whereas the left side grows by less than one. Consequently

$$|S| = \ell(q - 2m_0) \ge \frac{hq}{4}. \tag{10}$$

3.3 Cubic completion and every sufficiently large order

The following operation will preserve the target antichain.

Lemma 5. *Let B be a connected subcubic graph with a degree-two root y, exactly k leaves, and a cover by k geodesics from y to those leaves. There is a connected cubic graph $C(B)$ of order $2|V(B)|+4k$ containing an isometric copy of B and satisfying $p_{y^0}(C(B)) \le 3k$.*

Proof. Take copies B^0, B^1. At every degree-two vertex add v^0v^1. At a leaf u, add a fresh four-cycle $c_0c_1c_2c_3c_0$, joining u^0 to c_0, c_1 and u^1 to c_2, c_3. Every vertex now has degree three, and y^0y^1 connects the copies. Projection to B, collapsing each cycle to its leaf, sends edges to edges or stationary steps. Thus each old copy is isometric.

Write $\lambda(v) = d_B(y, v)$. Distances from y^0 to v^0, v^1 are $\lambda(v), \lambda(v)+1$. The first is isometry. For the second, a path must switch copies, introducing at least one stationary step under projection, while the root edge followed by a copy-one geodesic attains the bound. At a leaf u the four cycle distances are $\lambda(u)+1, \lambda(u)+1, \lambda(u)+2, \lambda(u)+2$. For each rooted leaf geodesic use three paths: its copy-zero version extended by c_0, c_3; its copy-zero version extended by c_1, c_2; and y^0y^1 followed by its copy-one version. The distance formulas make all three geodesics, and these $3k$ paths cover $C(B)$. □

Apply the lemma to $B^{(h)}$. Isometry preserves both observer distances and pairwise distances, and hence preserves S as a position set in copy zero. The resulting graph $C^{(h)}$ has

$$n_h = 2N_h + 4k = 2wL + 8k - 2 = \Theta(h2^h), \qquad \rho(C^{(h)}) \ge \frac{\ell(q-2m_0)}{3k} = \Omega(2^h). \tag{11}$$

The formulas for N_h and n_h increase with h; their two-sided $\Theta(h2^h)$ estimates give bounded consecutive ratios. Extending a single ray beyond all its attachments by t vertices leaves old distances, the number of leaves, and the size of the rooted geodesic cover unchanged. It adds t vertices to the base and $2t$ vertices to its cubic completion. Given any sufficiently large n, choose the largest base order $N_h \le n$ and pad to n. For an even n, choose the largest cubic order $n_h \le n$ and pad the base by $(n-n_h)/2$. Bounded consecutive ratios keep the number of targets proportional to n, while the cover size is $O(\log n)$. This proves the lower bounds in the first two lines of Theorem 1; Corollary 4 proves their upper bounds.

For instance, $h = 10$ gives $q = 1024$, $d = 3$, $\ell = 8$, $w = 25$, $k = 32$, $M = 15$, $t_0 = 32$, and $L = 8344$. Thus

$$|V(C^{(10)})| = 417454, \qquad \mathrm{vp}(C^{(10)}) \ge 8160, \qquad \mathrm{vp}^-(C^{(10)}) \le 96, \qquad \rho(C^{(10)}) \ge 85. \tag{12}$$

3.4 Maximum degree exactly Δ

Thankachy et al. [1, Theorem 2.9] prove $a \ge \lceil(\Delta+1)/3\rceil$. Indeed, from any observer the closed neighborhood of a maximum-degree vertex lies in three consecutive distance layers; one is large enough to give the bound (if the observer is that vertex, use its neighborhood). Together with (2), this gives

$$\rho(G) \le \min\left\{\frac{3n}{\Delta+1}, \frac{4n}{\log n}\right\} \le \frac{7n}{\Delta+1+\log n}.$$

The final inequality follows by adding the two resulting lower bounds on $1/\rho$ with weights $3/7$ and $4/7$.

For $3 \le \Delta \le n/2$, take a padded base construction of order $N = n-(\Delta-2) \ge n/2$ and attach $\Delta-2$ new leaves at its degree-two root. Its maximum degree is exactly Δ, old distances are preserved, and its root cover has $k+\Delta-2 = O(\log n+\Delta)$ paths. The old target set has size $\Omega(N) = \Omega(n)$, with absolute constants. For $\Delta > n/2$, the quantity $n/(\Delta+\log n)$ is

bounded above and below by absolute positive constants. A star with Δ leaves, with one edge subdivided to obtain order n, has the required degree and ratio at least one. This completes the degree statement.

4 Realizable pairs

If $p_y = 1$, Lemma 2 gives a spanning geodesic. Any extra edge would shortcut it, so the graph is a path. Conversely, a path endpoint has value one. Thus the first row consists of $(1, 1)$, attained by K_1, K_2, and $(1, 2)$, attained by every longer path, as in [1, Corollary 2.7].

4.1 The exact second row

Proposition 6. *If* $\mathrm{vp}^-(G) = 2$, *then* $\mathrm{vp}(G) \le 10$.

Proof. Choose y with $p_y = 2$ and cover the graph by two y-geodesics P, Q, indexed by distance from y. They may share vertices. Put the observer at $x = P_r$, interchanging the paths if necessary. Since P is isometric, $d(x, P_i) = |i - r|$, so an x-position set contains at most two vertices of P.

A contact is a shared vertex $P_i = Q_j$, of cost $\epsilon = 0$, or an edge P_iQ_j, of cost $\epsilon = 1$. Shared vertices have $i = j$ and edges have $|i - j| \le 1$. Taking the last visit to P in a shortest path gives

$$d(x, Q_t) = \min_{(i,j,\epsilon)} \{|i - r| + \epsilon + |t - j|\}. \tag{13}$$

After its last visit the path lies in Q; no chord between nonconsecutive vertices of Q is possible because Q is isometric. Conversely every expression in (13) is the length of a walk. Write

$$|i - r| + \epsilon = |j - r| + c, \qquad c \in \{0, 1, 2\}.$$

Separate contacts into $j \le r$ and $j \ge r$, assigning equality to one group, and let f_-, f_+ be their lower envelopes. For a nonempty left group let J be its largest contact index. When $t \le J$,

$$f_-(t) = r - t + g(t), \qquad g(t) \in \{0, 1, 2\}.$$

The lower bound is termwise and the contact at J gives the upper bound. Two selected vertices assigned to this envelope with the same g-value are comparable along Q. There are at most three such selected vertices with $t \le J$. For $t > J$ all terms have slope $+1$, contributing at most one more. Reflection gives at most four for the right group as well. Assign each selected vertex of $Q \setminus P$ to an envelope attaining its actual distance; the comparisons therefore use actual distances. There are at most eight selected vertices outside P, hence at most ten in all. □

For explicit realizations we also need a bound that counts contacts.

Lemma 7. *A graph made of two paths sharing just their root and s additional rungs, each joining positive levels differing by at most one, satisfies $p_x \le s + 4$ for every observer x.*

Proof. Use (13) with $x = P_r$. Suppose q rungs are assigned to the left group. If $q > 0$, any one dominates the root contact on positive Q-indices, because its offset is at most two and its level is at least one. For these rungs

$$g(t) = f_-(t) + t - r = \min_j \{c_j + 2\max(t - j, 0)\}.$$

Up to their last contact index J, $g(t) \in \{0, 1, 2\}$ and every attained value is one of the offsets c_j: values zero and one require that offset; if no offset is two, the last contact bounds $g \le 1$

throughout this range. There are thus at most q selected vertices before or at J, and one in the final slope-+1 tail. If $q = 0$, the root contact contributes a single chain. The reflected right group contributes at most its number of rungs plus one (or zero if it is empty). The other ray therefore contributes at most $s + 2$, and the observer's ray at most two. □

Take two length-13 paths with common root $P_0 = Q_0 = y$ and define

$$e_1 = P_4Q_5, \quad e_2 = P_3Q_3, \quad e_3 = P_2Q_1,$$
$$e_4 = P_{10}Q_9, \quad e_5 = P_{11}Q_{11}, \quad e_6 = P_{12}Q_{13}.$$

The following table specifies five graphs and selected sets with observer $x = P_7$. The last column lists the corresponding x-distances.

b	Rung indices	Selected on P	Selected Q-indices	Q-distances
6	$1, 4$	y, P_{13}	$2, 6, 8, 13$	$7, 5, 5, 8$
7	$1, 2, 4$	y, P_{13}	$1, 4, 6, 8, 13$	$7, 5, 5, 5, 8$
8	$1, 2, 3, 4$	P_1, P_{13}	$1, 2, 4, 6, 8, 13$	$6, 6, 5, 5, 5, 8$
9	$1, 2, 3, 4, 5$	P_1, P_{13}	$1, 2, 4, 6, 8, 10, 13$	$6, 6, 5, 5, 5, 5, 7$
10	$1, 2, 3, 4, 5, 6$	P_1, P_{13}	$1, 2, 4, 6, 8, 10, 12, 13$	$6, 6, 5, 5, 5, 5, 6, 6$

These graphs are connected and subcubic, with 27 vertices and $s = b - 4$ rungs, so Lemma 7 gives $\mathrm{vp} \le b$. Formula (13) gives the displayed distances. Successive selected Q-indices have absolute distance-value difference strictly smaller than their index difference. By the triangle inequality the same is true for any pair, so the selected Q-vertices are incomparable. The selected P_1, P_{13} have distance six, and y has distance seven. Cross-ray differences are at most one, except for y against a distance-five vertex and P_1 or P_{13} against Q_{13} of distance eight. The former pairs have level separation at least four. In the latter cases $d(P_{13}, Q_{13}) = 8$, by the closest rung e_4, while P_1, Q_{13} have level separation twelve. All remaining pairs with distance difference one are nonadjacent. Thus all cross-ray comparisons are strict, and each displayed set is a position set of size b.

Both rays are still y-geodesics, so $p_y \le 2$. The graphs contain cycles, so the first-row characterization gives $\mathrm{vp}^- \ge 2$. They realize exactly $(2, b)$ for $6 \le b \le 10$. The next proposition supplies $b = 2, 3, 4, 5$; Proposition 6 excludes every larger value.

4.2 Intervals and exponential row growth

Proposition 8. *Every integer pair $a \ge 2$, $a \le b \le 3a - 1$ is realizable.*

Proof. For $b = a$ use K_{a+1}. Otherwise choose $1 \le r, t \le a$ with $r + t = b + 1 - a$; this is possible because $2 \le b + 1 - a \le 2a$. Take a vertex y, independent sets A, D of sizes r, t, and a clique B of size a. Add all edges from y to A, from A to B, and from B to D, with no others. The graph has $b + 2$ vertices. Its position numbers are

$$p_y = a, \qquad p_u = \max\{a + 1, r + t - 1, t + 1\} \quad (u \in A),$$
$$p_u = \max\{a, r + t - 1\} \quad (u \in D), \qquad p_u = \max\{r + a - 1 + t, a + t\} = b \quad (u \in B).$$

For completeness, from y the layers are A, B, D, and every earlier-layer vertex is comparable with every later-layer vertex. From $u \in A$ the layers are $\{y\} \cup B$ and $(A \setminus \{u\}) \cup D$. A selected vertex of B forbids the second layer, whereas y forbids $A \setminus \{u\}$ but allows D. From $u \in D$ the layers are B, $A \cup (D \setminus \{u\})$, and $\{y\}$; a vertex of B forbids later layers, and y forbids A. From $u \in B$ the only nonneighbor is y, comparable precisely with A. These observations give both the upper bounds and attaining choices in the displayed formulas. All values lie between a and b, with both attained. □

The known pairs $(a, 6a-4)$ for $a \geq 3$ are due to [1, Lemma 2.3]. The exponential lower bound in (1) has an exact lower parameter, as follows. Set $h = a-1$, take a rays $P_0, \ldots, P_h$ attached directly to a common root y, and use only the dyadic edges (4). Put

$$A = 4(2^h - 1), \quad t_0 = h + 2, \quad s = A + t_0, \quad F = A + h,$$

and give all rays length $L > s + A$. Define C_i as before, and let $x = P_{0,s}$. The potential equal to $F + 1$ at y and $\min\{F + 1, F_i + \mathrm{dist}(t, C_i)\}$ on ray i is 1-Lipschitz: at its first vertex the untruncated value is $s + i - 1 = F + 1 + i$. The paths used in (8) show that the 2^h centers on P_h all have x-distance F.

Attach one further leaf z at y. The graph has $a + 1$ leaves, so every observer can select at least a other leaves. On the other hand the a paths from z through y along the rays are geodesics and cover the graph; their lengths are certified by the original level function shifted by one. Thus $p_z = a$ and $\mathrm{vp}^- = a$. The old target sphere is preserved. This proves $B(a) \geq 2^{a-1}$, while (2) proves the upper bound and the existence of the maximum, since each row is nonempty and bounded in the integers. Therefore $B(a) = 2^{\Theta(a)}$.

5 Local monophonic position and vertex visibility

An induced path is also called a monophonic path. For $|V(G)| \geq 2$, let m_x be the largest size of $S \subseteq V(G) \setminus \{x\}$ such that no induced x, v-path contains another vertex of S, for any $v \in S$. This localizes the universal induced-path condition of monophonic position [5]. For the fan $K_1 \vee P_n$ with universal vertex w, $m_w = n$: every induced path from w to a path vertex is its edge. At a path vertex at most one selected path vertex lies on either side; the only other candidate is w. Its neighborhood is an admissible set, so path endpoints have value two and internal path vertices have value three. Hence $\min_x m_x = 2$, $\max_x m_x = n$ for $n \geq 2$.

The existential shortest-path version already has standard terminology. Following Roy–Di Stefano–Klavžar–Lakshmanan [3], an x-visibility set is a set $S \subseteq V(G) \setminus \{x\}$ for which every $v \in S$ has some shortest x, v-path meeting S just at v. Its largest size is $v_x(G)$, and $\mathrm{vv}(G) = \max_x v_x(G)$ is the vertex visibility number. We write $\mathrm{vv}^-(G) = \min_x v_x(G)$ for the minimum over observers. This condition does not impose mutual visibility between two selected vertices.

Proposition 9. *There is a connected cubic graph D_t of order $4t+2$, for every $t \geq 1$, such that $\mathrm{vv}(D_t)/\mathrm{vv}^-(D_t) \geq t/7$. Thus vertex-visibility ratios have linear growth along a cubic family.*

Proof. Start with two length-t paths P, Q sharing only their root y and add the rungs P_iQ_i for $1 \leq i \leq t$. Call the graph L_t. Its root geodesics are the unique ray paths, so $v_y(L_t) = 2$. From $x = P_1$, the set $\{Q_1, \ldots, Q_t\}$ is visible: the paths $P_1P_2 \cdots P_iQ_i$ have length i, are shortest, and contain no other selected vertex. The lower bound on length follows either by using a rung (at least $i - 1$ level-changing steps and one rung) or by going through y (at least $i + 1$ steps).

Take two copies of L_t and join corresponding vertices y, P_t, Q_t. These are exactly the degree-two vertices, so the result D_t is cubic of order $4t + 2$ and is connected. Projection makes either copy isometric, so the same target set is visible from P_1^0. From y^0, the geodesics to P_i^0, Q_i^0 are unique. For $i < t$, those to P_i^1, Q_i^1 are also unique: the root edge followed by the ray has length $i + 1$, whereas entering through an end edge costs at least $2t - i + 1 > i + 1$. No rung can occur on these root geodesics because it does not increase level.

Partition all nonobserver vertices into the four ray chains $\{P_i^0\}_{1 \leq i \leq t}$, $\{Q_i^0\}_{1 \leq i \leq t}$, $\{P_i^1\}_{1 \leq i < t}$, $\{Q_i^1\}_{1 \leq i < t}$, and the three singletons y^1, P_t^1, Q_t^1. A visibility set meets each unique-geodesic chain in at most one vertex. Hence $v_{y^0}(D_t) \leq 7$ and the claimed ratio follows. The general upper bound $\mathrm{vv} / \mathrm{vv}^- \leq |V(G)| - 1$ gives the matching order along this family. □

6 Exact checks and the remaining pair problem

The proofs above establish the all-parameter statements. Exact finite checks provide separate verification of the constructions. For each observer, the geodesic order gives

$$p_x(G) = |V(G)| - 1 - \nu_x,$$

where ν_x is the maximum matching size in the bipartite graph of all strict comparable pairs. Indeed a chain partition gives such a matching by linking successive vertices, and matching edges concatenate into disjoint chains because strict order has no directed cycles; Dilworth's theorem identifies their minimum number with the maximum antichain size. All strict comparable pairs are necessary in this reduction, not just the edges of G. Polynomial-time computability was established in [1, Theorem 5.7 and Corollary 5.8].

Matching and an independent recursive maximum-antichain search agree on every observer in the five 27-vertex witnesses, all 1,834 two-ray graphs with each ray of length one through four and every subset of allowed rungs, and all 27,476 connected labeled graphs of orders one through six (164,031 observer values in the last collection). The two-ray upper bounds also hold in 400 further sampled graphs with ray lengths five through fifteen. The nine-vertex graph with edges

$$\{05, 06, 07, 15, 17, 26, 28, 37, 38, 48, 57, 68\}$$

has the independently recomputed vector $(4, 2, 3, 6, 3, 3, 5, 4, 4)$ and hence pair $(2, 6)$.

For the large constructions, breadth-first searches verify every potential edge inequality, every specified center distance, every quotient distance at a selected target, and every rooted covering path for $h = 8, 9, 10$, as well as a padded $h = 8$ example. Antichain verification uses the shortest-path directed acyclic graph: propagate a flag forward from selected vertices along distance-increasing edges and reject any later selected vertex reached by a flag. This checks all pairs implicitly in linear time, including targets of different radii. The cubic graphs have orders 101454, 204158, 417454 and 101488 in these four checks. The exponential-row constructions are checked for $2 \le a \le 10$, all interval examples for $2 \le a \le 15$, and the visibility construction for $1 \le t \le 12$ and $t = 20, 50, 100$. These finite ranges support the proofs and make no claim of minimum order for witnesses.

The higher pair rows remain open: determine $B(a)$ exactly for $a \ge 3$, and decide whether every integer between a and $B(a)$ is realizable. Neither the exponential bounds nor the interval construction settles these questions. Sharp leading constants in the cubic and exact-degree growth estimates are also not determined here.

References

[1] M. G. S. Thankachy, E. J. Thomas, U. Chandran S. V., J. Tuite, G. Di Stefano and G. Erskine, *On the Vertex Position Number of Graphs*, arXiv:2209.00359v2 (2022). Numbered statements cited here refer to this version.

[2] M. Dumas, F. Foucaud, A. Perez and I. Todinca, *On graphs coverable by k shortest paths*, arXiv:2206.15088v2 (2023).

[3] D. Roy, G. Di Stefano, S. Klavžar and A. Lakshmanan S, *The vertex visibility number of graphs*, arXiv:2510.19452v1 (2025).

[4] R. P. Dilworth, *A decomposition theorem for partially ordered sets*, Ann. of Math. (2) **51** (1950), 161–166.

[5] E. J. Thomas, U. Chandran S. V., J. Tuite and G. Di Stefano, *On monophonic position sets in graphs*, arXiv:2012.10330v4.